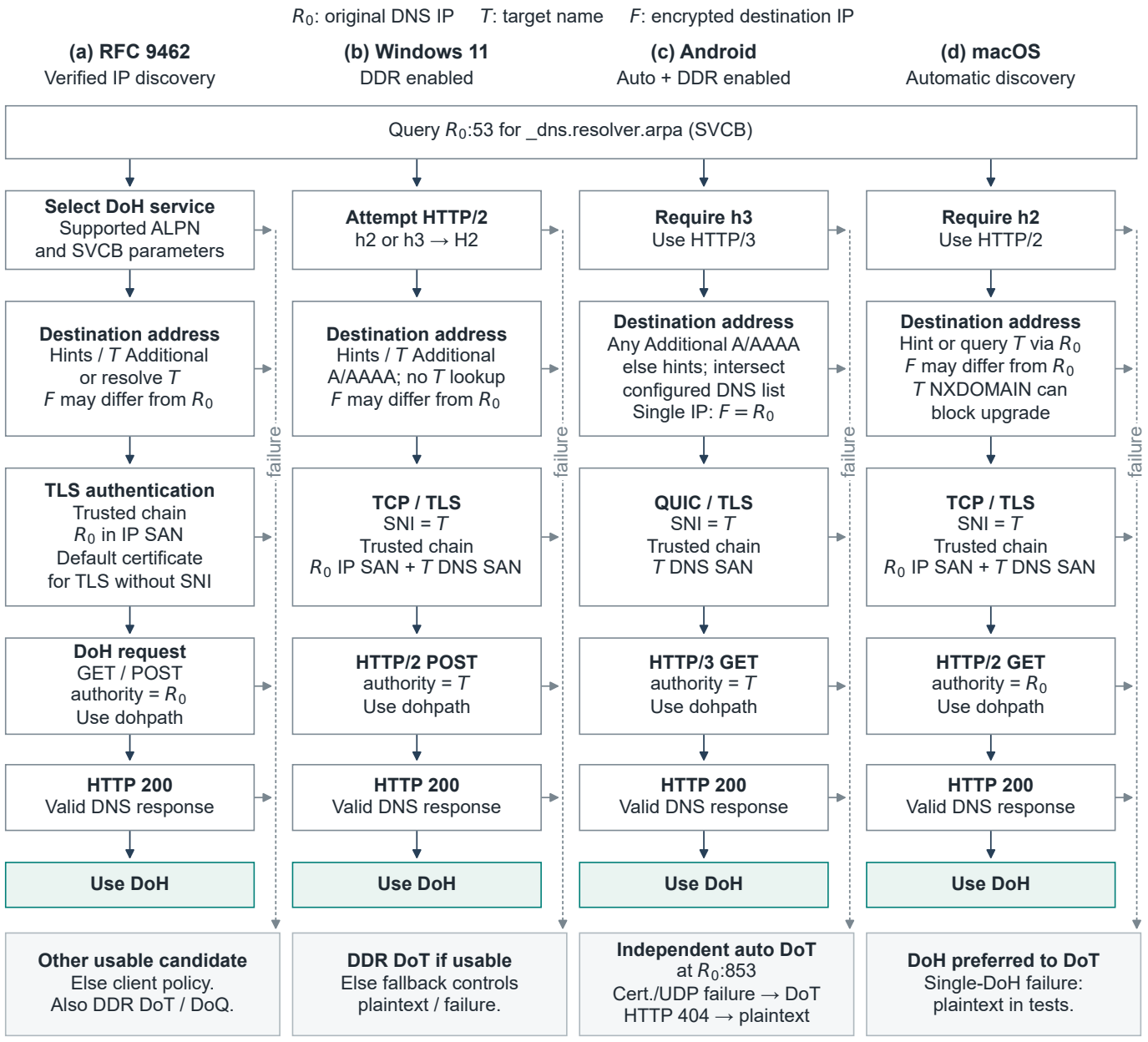

R_0: original DNS IP T: target name F: encrypted destination IP
(a) RFC 9462
Verified IP discovery
(b) Windows 11
DDR enabled
(c) Android
Auto + DDR enabled
(d) macOS
Automatic discovery
Query R_0:53 for _dns.resolver.arpa (SVCB)
Select DoH service
Supported ALPN and SVCB parameters
Attempt HTTP/2
h2 or h3 → H2
Require h3
Use HTTP/3
Require h2
Use HTTP/2
Destination address
Hints / T Additional or resolve T
F may differ from R_0
Destination address
Hints / T Additional A/AAAA; no T lookup
F may differ from R_0
Destination address
Any Additional A/AAAA else hints; intersect configured DNS list
Single IP: $F = R_0$
Destination address
Hint or query T via R_0
F may differ from R_0
T NXDOMAIN can block upgrade
TLS authentication
Trusted chain
R_0 in IP SAN
Default certificate for TLS without SNI
TCP / TLS
SNI = T
Trusted chain
R_0 IP SAN + T DNS SAN
QUIC / TLS
SNI = T
Trusted chain
T DNS SAN
TCP / TLS
SNI = T
Trusted chain
R_0 IP SAN + T DNS SAN
DoH request
GET / POST
authority = R_0
Use dohpath
HTTP/2 POST
authority = T
Use dohpath
HTTP/3 GET
authority = T
Use dohpath
HTTP/2 GET
authority = R_0
Use dohpath
HTTP 200
Valid DNS response
HTTP 200
Valid DNS response
HTTP 200
Valid DNS response
HTTP 200
Valid DNS response
Use DoH
Use DoH
Use DoH
Use DoH
failure
failure
failure
failure
Other usable candidate
Else client policy.
Also DDR DoT / DoQ.
DDR DoT if usable
Else fallback controls plaintext / failure.
Independent auto DoT
at R_0:853
Cert./UDP failure → DoT
HTTP 404 → plaintext
DoH preferred to DoT
Single-DoH failure: plaintext in tests.